Logistic Regression classification Report

Preprocessing = get_dummies, StandardScaler

Cm = [[51 0]
[[1 81]]

	precision	recall	f1-score	support
0	0.98	1.00	0.99	51
1	1.00	0.99	0.99	82
accuracy			0.99	133
macro avg	0.99	0.99	0.99	133
weighted avg	0.99	0.99	0.99	133

- The f1_macro value for best parameter {'C': 1.0, 'class_weight': None, 'dual': False, 'fit_intercept': True, 'intercept_scaling': 1, 'l1_ratio': 0.0, 'max_iter': 100, 'n_jobs': None, 'penalty': 'l2', 'random_state': None, 'solver': 'sag', 'tol': 0.0001, 'verbose': 0, 'warm_start': False}. 0.9924946382275899

- value for roc_auc_score = 1.0

KNN Classification Report

Preprocessing = get_dummies, StandardScaler

Cm = [[51 0]
[8 74]]

	precision	recall	f1-score	support
0	0.86	1.00	0.93	51
1	1.00	0.90	0.95	82
accuracy			0.94	133
macro avg	0.93	0.95	0.94	133
weighted avg	0.95	0.94	0.94	133

- The f1_macro value for best parameter {'algorithm': 'auto', 'leaf_size': 30, 'metric': 'minkowski', 'metric_params': None, 'n_jobs': None, 'n_neighbors': 5, 'p': 2, 'weights': 'distance'}. 0.940494593126172

- The value for roc_auc_score = 1.0

Gaussian Naïve Bayes

Preprocessing = get_dummies, StandardScaler

Cm = [[51 0]
 [3 79]]

precision	recall	f1-score	support	
0	0.94	1.00	0.97	51
1	1.00	0.96	0.98	82
accuracy			0.98	133
macro avg	0.97	0.98	0.98	133
weighted avg	0.98	0.98	0.98	133

- The f1_macro value for best parameter {'priors': None, 'var_smoothing': 1e-09} . 0.9775556904684072

- The value for roc_auc_score = 1.0

Multinomial Naïve Bayes

Preprocessing = get_dummies, MiniMax Scaler

```
Cm = [[51  0]
      [ 2 80]]
```

	precision	recall	f1-score	support
0	0.96	1.00	0.98	51
1	1.00	0.98	0.99	82
accuracy			0.98	133
macro avg	0.98	0.99	0.98	133
weighted avg	0.99	0.98	0.99	133

- The f1_macro value for best parameter {'alpha': 1.0, 'class_prior': None, 'fit_prior': True, 'force_alpha': True}. 0.9850141736106648

- The value for roc_auc_score = 1.0

Complement Naïve Bayes

Preprocessing = get_dummies, MiniMax Scaler

```
Cm = [[51  0]
      [ 2 80]]
```

	precision	recall	f1-score	support
0	0.96	1.00	0.98	51
1	1.00	0.98	0.99	82
accuracy			0.98	133
macro avg	0.98	0.99	0.98	133
weighted avg	0.99	0.98	0.99	133

- `The f1_macro value for best parameter {'alpha': 1.0, 'class_prior': None, 'fit_prior': True, 'force_alpha': True, 'norm': False}. 0.9850141736106648`

- The value for roc_auc_score = 1.0

Bernouli NB

Preprocessing = get_dummies, MiniMax Scaler

```
Cm = [[51  0]
      [ 7 75]]
```

	precision	recall	f1-score	support
0	0.88	1.00	0.94	51
1	1.00	0.91	0.96	82
accuracy			0.95	133
macro avg	0.94	0.96	0.95	133
weighted avg	0.95	0.95	0.95	133

- `The f1_macro value for best parameter {'alpha': 1.0, 'binarize': 0.0, 'class_prior': None, 'fit_prior': True, 'force_alpha': True}. 0.9478851104269762`

- The value for roc_auc_score `0.9966523194643712`

Categorical NB

Preprocessing = get_dummies, MiniMax Scaler

```
Cm = [[51  0]
      [ 5 77]]
```

	precision	recall	f1-score	support
0	0.91	1.00	0.95	51
1	1.00	0.94	0.97	82
accuracy			0.96	133
macro avg	0.96	0.97	0.96	133
weighted avg	0.97	0.96	0.96	133

- The f1_macro value for best parameter {'alpha': 1.0, 'class_prior': None, 'fit_prior': True, 'force_alpha': True, 'min_categories': None}. 0.9626932787797391

- The value for roc_auc_score is 0.996771879483500

Random Forest classification

Preprocessing = get_dummies, StandardScaler

```
Cm = [[50  1]
      [ 1 81]]
```

	precision	recall	f1-score	support
0	0.98	0.98	0.98	51
1	0.99	0.99	0.99	82
accuracy			0.98	133
macro avg	0.98	0.98	0.98	133
weighted avg	0.98	0.98	0.98	133

The f1_macro value for best parameter: {'bootstrap': True, 'ccp_alpha': 0.0, 'class_weight': None, 'criterion': 'log_loss', 'max_depth': None, 'max_features': 'log2', 'max_leaf_nodes': None, 'max_samples': None, 'min_impurity_decrease': 0.0, 'min_samples_leaf': 1, 'min_samples_split': 2, 'min_weight_fraction_leaf': 0.0, 'monotonic_cst': None, 'n_estimators': 100, 'n_jobs': None, 'oob_score': False, 'random_state': None, 'verbose': 0, 'warm_start': False}. 0.9849624060150376

The value of roc_auc_score is 0.9997608799617408

Decision Tree Classification

Preprocessing = get_dummies, StandardScaler

```
Cm = [[51  0]
      [ 3 79]]
```

precision	recall	f1-score	support	
0	0.94	1.00	0.97	51
1	1.00	0.96	0.98	82
accuracy			0.98	133
macro avg	0.97	0.98	0.98	133
weighted avg	0.98	0.98	0.98	133

- The f1_macro value for best parameter {'ccp_alpha': 0.0, 'class_weight': None, 'criterion': 'log_loss', 'max_depth': None, 'max_features': None, 'max_leaf_nodes': None, 'min_impurity_decrease': 0.0, 'min_samples_leaf': 1, 'min_samples_split': 2, 'min_weight_fraction_leaf': 0.0, 'monotonic_cst': None, 'random_state': None, 'splitter': 'random'}. 0.9775556904684072

- The value for roc_auc_score is 0.9817073170731707

Support Vector Machine Classification

Preprocessing = get_dummies, StandardScaler

```
Cm = [[51  0]
      [ 1 81]]
```

precision	recall	f1-score	support	
0	0.98	1.00	0.99	51
1	1.00	0.99	0.99	82
accuracy			0.99	133
macro avg	0.99	0.99	0.99	133
weighted avg	0.99	0.99	0.99	133

- The f1_macro value for best parameter {'C': 1.0, 'break_ties': False, 'cache_size': 200, 'class_weight': None, 'coef0': 0.0, 'decision_function_shape': 'ovr', 'degree': 3, 'gamma': 'scale', 'kernel': 'rbf', 'max_iter': -1, 'probability': False, 'random_state': None, 'shrinking': True, 'tol': 0.001, 'verbose': False}. 0.9924946382275899

- The value for roc_auc_score is 1.0

SGD Classifier

Preprocessing = get_dummies, StandardScaler

```
Cm = [[50  1]
      [ 0 82]]
```

precision	recall	f1-score	support	
0	1.00	0.98	0.99	51
1	0.99	1.00	0.99	82
accuracy			0.99	133
macro avg	0.99	0.99	0.99	133
weighted avg	0.99	0.99	0.99	133

- The f1_macro value for best parameter {'alpha': 0.0001, 'average': False, 'class_weight': None, 'early_stopping': False, 'epsilon': 0.1, 'eta0': 0.01, 'fit_intercept': True, 'l1_ratio': 0.15, 'learning_rate': 'optimal', 'loss': 'log_loss', 'max_iter': 1000, 'n_iter_no_change': 5, 'n_jobs': None, 'penalty': 'l2', 'power_t': 0.5, 'random_state': None, 'shuffle': True, 'tol': 0.001, 'validation_fraction': 0.1, 'verbose': 0, 'warm_start': False}. 0.9924667654735397

- The roc_auc_score is 0.9997608799617408

Final model:

Out of 11 model, **The Logistic regression model** achieved an overall accuracy of 99%, meaning it correctly classified 99% of the test data **and ROC_AUC value = 1.0**. For Class 0 (No), the precision is 98%, recall is 1.0, and F1-score is 99%, showing excellent performance. For Class 1 (Yes), the precision is 1.00%, recall is 99%, and F1-score is 99%, indicating good performance, although the model misses some customers who actually purchased. Overall, the weighted F1-score is 99%, which shows the model performs well across both classes.